

SOME REMARKS ON THE GRUBBS ESTIMATOR

F. Lombard and C.J. Potgieter

Department of Statistics, University of Johannesburg
and

Institute for Applied Mathematics and Computational Science, Texas A & M
University

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Summary: We consider estimation of the precision of a measuring instrument without the benefit of replicate observations on heterogeneous sampling units. Grubbs (1948) proposed an estimator which involves the use of two additional measuring instruments, resulting in a triple of observations on each sampling unit. Since the precisions of the measuring instruments are generally different, these observations cannot be treated as replicates. Very large sample sizes are often required if the standard error of the estimate is to be within reasonable bounds and if negative precision estimates are to be avoided.

The Grubbs estimator is not the maximum likelihood estimator (MLE). We show that the efficiency of the Grubbs estimator relative to the MLE is determined largely by the extent of sampling unit heterogeneity. Our results are presented in the context of the evaluation of on-line analysers. Two data sets from actual analyser evaluations are used to illustrate the results.

1. Introduction

The problem considered in this paper, first discussed by Grubbs (1948, 1973), involves estimation of the precision of a measuring instrument without the benefit of replicate observations on the sampling units. In our applications, the instrument is an on-line analyser that measures coal quality characteristics
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in real time, as opposed to the usual rather time consuming method consisting of mechanical sampling from a coal stream followed by chemical analysis in a laboratory. In these applications replicate measurements cannot be made because coal on a conveyor belt can pass through the analyser only once. Furthermore, there is substantial fluctuation in the average daily values of the measured characteristic.

Nonetheless, some form of repeated measurement is possible, namely that the characteristic of interest (ash content of the coal, for instance) can be measured simultaneously by two additional instruments. In the coal analyser setup it is possible to obtain every day three samples of coal that had passed through the analyser (instrument 1 which gives a reading Y), one sample coming from a mechanical sampler (instrument 2) and another by manual sampling that involves stopping the conveyor belt at regular time intervals (instrument 3). The latter two samples are then submitted for chemical analysis, yielding respective readings X_1 and X_2 . The main focus of interest is in assessing the precision of the analyser. If n independent sets of readings involving n daily coal samples are obtained, the data consist of a $n \times 3$ matrix with j -th row (Y_j, X_{j1}, X_{j2}) , the n rows being statistically independent.

An underlying structure is assumed, namely

$$Y_j = T_j + e_j, \quad X_{1j} = \mu_1 + T_j + e_{1j}, \quad X_{2j} = \mu_2 + T_j + e_{2j} \quad (1)$$

where T_j has mean ν and variance τ^2 , μ_1 and μ_2 are fixed (nonrandom, but unknown) numbers and where the measurement errors e_j , e_{1j} and e_{2j} are independently and normally distributed with zero means and variances σ^2 , σ_1^2 and σ_2^2 respectively. The random variable T_j represents the numerical value of the coal characteristic that is measured with error and which varies

from sample to sample. The variance of T_j , denoted by τ^2 , is commonly referred to as the “product variance”. The parameters μ_1 and μ_2 represent unknown measurement biases while the random variables e_j , e_{1j} and e_{2j} represent measurement errors. The statistical problem is to find a confidence interval with specified confidence coefficient for σ^2 , hence for the standard deviation σ .

Grubbs (1948) proposed an estimator that can be expressed as

$$\tilde{\sigma}^2 = \max\{0, S_{Y-X_1, Y-X_2}\} \quad (2)$$

where $S_{Y-X_1, Y-X_2}$ denotes the sample covariance between the data vectors $Y - X_1$ and $Y - X_2$. This is a non-parametric estimator which is *asymptotically* normally distributed with mean σ^2 and variance

$$\text{var}(\tilde{\sigma}^2)^{-1} \{2\sigma^4 + \sigma^2(\sigma_1^2 + \sigma_2^2) + \sigma_1^2\sigma_2^2\}. \quad (3)$$

The paper by Rose (1991) can fairly be said to have led to the establishing of “Grubbs method” as the preferred methodology in the coal industry for estimating analyser precision – the methodology has been incorporated into at least one international standard – see International Standards Organization 15239 (2005). Substantial generalisations of the methodology have been made since Grubbs original work. In this regard, the work of Jaech (1985) and Christensen and Blackwood (1993) are particularly noteworthy. The latter works focus for the most part on questions of hypothesis testing, such as whether the precisions of multiple instruments may be regarded as being identical or not or whether one or more of a number of instruments are biased relative to the others. However, the evaluation of an online analyser presents questions that cannot easily be answered within a hypothesis testing context.

First, we typically know *a priori* that the analyser delivers greater precision than a process consisting of mechanical sampling followed by chemical

analysis. The question is rather whether the increase in precision is of an order that would justify the cost of acquiring and maintaining such an instrument. Second, bias in the readings produced by an analyser is typically not a problem because the evaluation of an online analyser usually consists of two phases: calibration followed by an assessment of precision. If an analyser cannot be calibrated to produce results that are unbiased relative to those produced by the reference instrument(s), the evaluation process terminates.

For our first data set we find $\tilde{\sigma}^2 = 0.013$, $\tilde{\sigma}_1^2 = 0.101$ and $\tilde{\sigma}_2^2 = 0.105$. The estimated standard error of the bias corrected (via the bootstrap) analyser standard deviation estimate $\tilde{\sigma} = 0.0066$ is 0.012. Given the cost involved in using two reference instruments, it is somewhat disappointing that 94 triples of observations could not deliver a standard error substantially smaller than 0.012. Notice, however, that the best among the two reference instruments ostensibly gives considerably less precise readings than the analyser so that it is hardly reasonable to expect a precise estimate of σ to arise from a comparison of the analyser with these reference instruments. This problem is inherent in the use of the Grubbs estimator in the present context.

Next, we observe that (2) is not the maximum likelihood estimator (MLE) based on the full set of data. Thus, it is an open question how much is lost by using (2) rather than the MLE. The main objective of the present paper is to investigate this question. We will show that the relative efficiency of the Grubbs estimator (2) relative to the MLE is determined largely by the extent of product variability, τ^2 . The larger (smaller) τ^2 is, the less (more) is the loss of efficiency.

In Section 2 we discuss briefly the bias characteristics of the Grubbs estimator as well as a difficulty inherent in using it to obtain a confidence interval for σ^2 . Section 3 is devoted to a comparative asymptotic analysis

of the Grubbs estimator and the MLE. In Section 4 we propose and discuss the use of a profile likelihood to find valid confidence intervals. We illustrate the proposed methodology using two sets of data (available from the authors upon request) collected in the evaluation of online analysers. Section 5 is devoted to examining the assumptions on which the proposed method rests and in Section 6 we summarise our main results. Some technical results are collected in the Appendix, Section 7.

We believe the Grubbs method's widespread use, especially in the coal industry, makes the method a relevant topic for research despite the fact that the methodology was first introduced more than sixty years ago.

2. Bias of the Grubbs estimator

Set

$$r_i^2 = \sigma_i^2 / \sigma^2, \quad i = 1, 2$$

and

$$v^2 = 2 + r_1^2 + r_2^2 + r_1^2 r_2^2. \quad (4)$$

By the central limit theorem, $S_{Y-X_1, Y-X_2}$ is approximately normally distributed with mean σ^2 and variance $\sigma^4 v^2 / n$. Assuming exact normality for simplicity, then a straightforward calculation shows for $\tilde{\sigma}^2$ in (2) that

$$E[\tilde{\sigma}^2] = \sigma^2 + B$$

where

$$B = \sigma^2 \left\{ \phi \left(n^{1/2} / v \right) v / n^{1/2} - \Phi \left(-n^{1/2} / v \right) \right\}. \quad (5)$$

Furthermore, the (approximate) probability that $S_{Y-X_1, Y-X_2}$ is negative is $\Phi \left(-n^{1/2} / v \right)$.

Table 1 shows the relative bias of $\tilde{\sigma}^2$, namely $B / \sigma^2 \times 100\%$, for $r = r_1 = r_2 = 1, 5, 10$ and $n = 30, 60, 90, 180$. The numbers in brackets

are the approximate probabilities that $S_{Y-X_1, Y-X_2}$ is negative, hence that the Grubbs estimator of σ^2 will be zero.

Table 1 Relative bias of the Grubbs estimator

	$r = 1$	$r = 5$	$r = 10$
$n = 30$	0.1 (0.01)	11.1 (0.18)	40.1 (0.31)
$n = 60$	0.0 (0.00)	3.8 (0.10)	20.3 (0.24)
$n = 90$	0.0 (0.00)	1.6 (0.06)	12.6 (0.20)
$n = 180$	0.0 (0.00)	1.4 (0.01)	4.5 (0.11)

Note the extent of the relative bias when r is large, as it is in our particular application. It is perhaps worthwhile pointing out that $\tilde{\sigma}^2$ is very nearly *median unbiased*. This follows from the fact that for $a > 0$, in view of the approximate normality of $S_{Y-X_1, Y-X_2}$,

$$Pr(\tilde{\sigma}^2 > a) \approx 1 - \Phi \left\{ n^{1/2}(a - \sigma^2)/\sigma^2 v \right\}$$

which equals 1/2 when $a = \sigma^2$.

The ultimate aim in an analysis of analyser evaluation data is to obtain for σ^2 a confidence interval having a prescribed coverage probability. If n is sufficiently large and if r_1 and r_2 , hence v , are *known* a $100(1 - \alpha)\%$ confidence interval is given by

$$(S_{Y-X_1, Y-X_2}/w_+, S_{Y-X_1, Y-X_2}/w_-)$$

where

$$w_{\pm} = 1 \pm z_{1-\alpha} v / n^{1/2}$$

and where $z_{1-\alpha}$ denotes the $100(1-\alpha)$ -th percentile of the standard normal distribution. In general, of course, r_1 and r_2 are not known and one can attempt to estimate them from the data. This approach often fails, especially when r_1 and r_2 are large. In the latter case $\tilde{\sigma}^2$ is frequently zero – see Table 1 – and this leads to a “confidence interval” consisting of a single point, namely $\sigma^2 = 0$. It would seem that the only manner in which a sensible confidence interval can be obtained without fail is if a likelihood method is used. This method is discussed and illustrated in Section 4.

3. Efficiency of the Grubbs estimator

The Grubbs estimator uses only the variables

$$U_1 = Y - X_1, U_2 = Y - X_2,$$

thereby eliminating the effect of the variable T , hence the product variance τ^2 , in (1). The covariance matrix of $\mathbf{U}^* = (U_1, U_2)$ is

$$\Omega_{\sigma^2, \psi} = \begin{bmatrix} \text{var}(U_1) & \text{cov}(U_1, U_2) \\ \cdot & \text{var}(U_2) \end{bmatrix} = \begin{bmatrix} \sigma^2 + \sigma_1^2 & \sigma^2 \\ \cdot & \sigma^2 + \sigma_2^2 \end{bmatrix}$$

and the likelihood on the data $\mathbf{U}_j^* = (U_{j1}, U_{j2})$, $j = 1, \dots, n$ (n independent realisations of $\mathbf{U}^*$) is

$$\mathcal{L}_2(\sigma^2, \psi, \nu) = \det(\Omega_{\sigma^2, \psi})^{-n/2} \exp \left\{ -\sum_{j=1}^n (\mathbf{U}_j^* - \nu) \Omega_{\sigma^2, \psi}^{-1} (\mathbf{U}_j^{*'} - \nu') / 2 \right\} \quad (6)$$

where $\psi = (\sigma_1^2, \sigma_2^2) \geq \mathbf{0}$, $\sigma^2 \geq 0$ and $\nu = (-\mu_1, -\mu_2)$ is unconstrained. It now follows from a result in Rao (1973, page 531) that the maximum likelihood estimator of $\sigma^2 = \text{cov}(U_1, U_2)$ is

$$\tilde{\sigma}^2 = \max\{0, S_{U_1, U_2}\} = \max\{0, S_{Y-X_1, Y-X_2}\},$$

which is precisely the expression in (2). Thus it is seen that the Grubbs estimator of σ^2 is, in fact, the MLE of σ^2 based on the *reduced data* $\mathbf{U}_j^* = (U_{j1}, U_{j2})$, $j = 1, \dots, n$.

To explore the relationship between the Grubbs estimator (i.e. the reduced data MLE) and the full data MLE, we introduce a third variable $U_3 = (X_1 + X_2)/2$. Clearly, the data sets (Y, X_1, X_2) and (U_1, U_2, U_3) are in one to one correspondence so that it is permissible to work with the latter representation of the data. The covariance matrix of (U_1, U_2, U_3) is

$$\Sigma_{\sigma^2, \theta} = \begin{bmatrix} \sigma^2 + \sigma_1^2 & \sigma^2 & -\sigma_1^2/2 \\ \cdot & \sigma^2 + \sigma_2^2 & -\sigma_2^2/2 \\ \cdot & \cdot & \tau^2 + (\sigma_1^2 + \sigma_2^2)/4 \end{bmatrix} \quad (7)$$

where $\theta = (\sigma_1^2, \sigma_2^2, \tau^2)$. Since $\text{var}(U_3) = \tau^2 + (\sigma_1^2 + \sigma_2^2)/4$ is functionally independent of σ^2 , it may seem at first sight that omitting the averages U_{j3} , $j = 1, \dots, n$ entails no loss of useful information about σ^2 and that attention may be confined to the Grubbs estimator of σ^2 . However, from (7), the linear correlation between U_i and U_3 ($i = 1, 2$), is

$$\rho_i = -\sigma_i^2 / \{(\sigma^2 + \sigma_i^2)(4\tau^2 + (\sigma_1^2 + \sigma_2^2))\}^{1/2},$$

which is nonzero and suggests that the averages U_{j3} , $j = 1, \dots, n$ do, in fact, contain “usable” information about σ^2 , especially when τ^2 is close to zero. Thus the MLE based on the full data set (U_{j1}, U_{j2}, U_{j3}) , $j = 1, \dots, n$ should, as expected, be more efficient than the MLE based on the reduced data (U_{j1}, U_{j2}) , $j = 1, \dots, n$. On the other hand, $\rho_i \rightarrow 0$ as $\tau^2 \rightarrow \infty$, which suggests in turn that the degree of superiority of the full data MLE should diminish as the product variance τ^2 increases. We now formalise these tentative conclusions.

Define in the usual way the asymptotic relative efficiency of the MLE with respect to Grubbs estimator for fixed σ^2 , σ_1^2 and σ_2^2 by

$$e_{\tau^2} = 100 \times \lim_{n \rightarrow \infty} \text{var}[n^{1/2}(\tilde{\sigma}^2 - \sigma^2)] / \text{var}[n^{1/2}(\hat{\sigma}^2 - \sigma^2)]$$

and express it as a percentage. In the Appendix we prove the following

Proposition 1 (i)

$$e_{\tau^2} \downarrow 100 \text{ as } \tau^2 \uparrow \infty. \quad (8)$$

(ii) As $\tau^2 \downarrow 0$ we have

$$\text{var}[n^{1/2}(\hat{\sigma}^2 - \sigma^2)] \rightarrow 2\sigma^2 \{ \sigma^4 + \sigma^2(\sigma_1^2 + \sigma_2^2) + \sigma_1^2\sigma_2^2/2 \} / (\sigma^2 + \sigma_1^2 + \sigma_2^2) \quad (9)$$

and

$$e_{\tau^2} \uparrow 100 \times (2+r_1+r_2+r_1r_2)(1+r_1+r_2) / (2+2r_1+2r_2+r_1r_2) > 100. \quad (10)$$

Thus, for “large” values of τ^2 , there is apparently little to choose between the Grubbs estimator and the full data MLE. For a numerical illustration, consider the following parameter configuration which is based on an analysis of our first set of coal analyser data:

$$\sigma^2 = 0.013, \sigma_1^2 = 0.101, \sigma_2^2 = 0.105, n = 94.$$

Figure 1 shows a plot of e_{τ^2} as a function of τ^2/σ^2 . The plot has been split into two segments, $0 < \tau^2/\sigma^2 \leq 40$ and $40 < \tau^2/\sigma^2 \leq 125$, each with its own vertical scale.

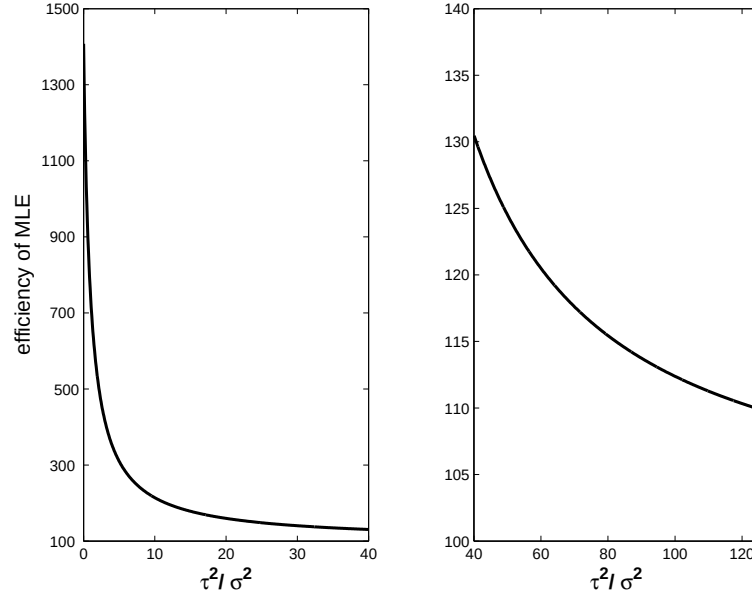


Figure 1. Efficiency of the MLE relative to the Grubbs estimator as a function of τ^2/σ^2 .

We find $e_0 = 1408\%$. Clearly, the Grubbs estimator can lose out to the full data MLE very badly if τ^2 is small. Furthermore, $e_{110} \approx 110\%$ which means in this very realistic case that even if τ^2 is two orders of magnitude greater than σ^2 there is still substantial gain if the full data MLE, rather than the Grubbs estimator, is used.

4. Confidence intervals

We now turn to the construction of a confidence interval for σ^2 . The preceding Section showed that the Grubbs estimator, which involves a reduction of the data, can be greatly inferior to the full data MLE in terms of variance. Thus, we consider a method that uses all the data. The likelihood function on the full

set of data

$$\mathbf{U}_j = (\mathbf{U}_j^*, U_{j3}) = (Y_j - X_{j1} Y_j - X_{j2} (X_{j1} + X_{j2})/2), \quad j = 1, \dots, n$$

is

$$\mathcal{L}(\sigma^2, \theta, \mu) = \det(\Sigma_{\sigma^2, \theta}^{-1})^{-n/2} \exp \left\{ -\sum_{j=1}^n (\mathbf{U}_j - \boldsymbol{\mu})' \Sigma_{\sigma^2, \theta}^{-1} (\mathbf{U}_j - \boldsymbol{\mu}) / 2 \right\} \quad (11)$$

where $\Sigma_{\sigma^2, \theta}$ is given in (7), $\theta = (\sigma_1^2, \sigma_2^2, \tau^2)$ and $\mu = (-\mu_1, -\mu_2, \nu + (\mu_1 + \mu_2)/2)$. Recall that $\nu = E[T]$, mean value of the product (coal in our applications) quality characteristic. Denote the maximum likelihood estimators of the parameters by $\hat{\sigma}^2$, $\hat{\theta}$ and $\hat{\mu}$. For the coal analyser data we find

$$\hat{\sigma}^2 = .0083, \quad \hat{\theta} = (.1046, .1077, 1.5659) \quad \text{and} \quad \hat{\mu} = (-0.01, -0.05, 26.13). \quad (12)$$

The profile likelihood for σ^2 ,

$$\mathcal{L}_{profile}(\sigma^2) = \mathcal{L}(\sigma^2, \hat{\theta}_{\sigma^2}, \hat{\mu}_{\sigma^2}) / \mathcal{L}(\hat{\sigma}^2, \hat{\theta}, \hat{\mu}),$$

where $\hat{\theta}_{\sigma^2}$ and $\hat{\mu}_{\sigma^2}$ denote the MLEs of θ and μ at the fixed value σ^2 , is plotted in Figure 2. If this were a normal likelihood, $-2\log \mathcal{L}_{profile}(\sigma^2)$ would have an (asymptotic) chi-squared distribution with 1 degree of freedom and an approximately 95% confidence interval for σ^2 would be

$$\left\{ \sigma^2 : \mathcal{L}_{profile}(\sigma^2) > \exp(-(1.96)^2/2) = .146 \right\} = (0, 0.025), \quad (13)$$

see Figure 2.

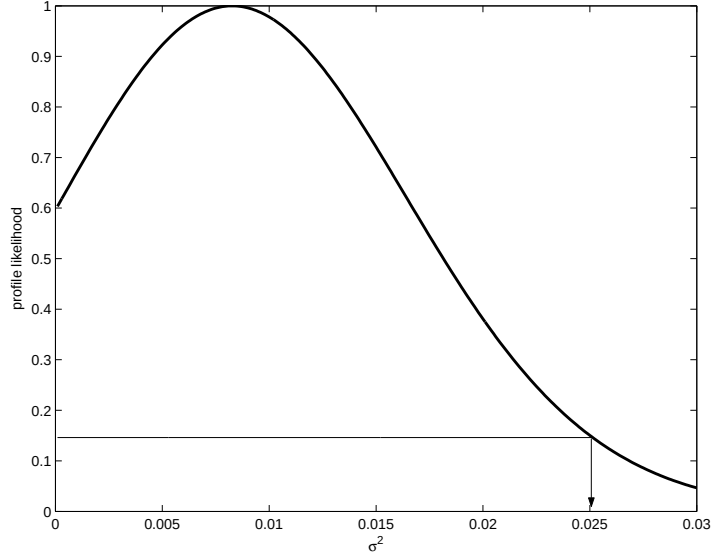


Figure 2. Profile likelihood of σ^2 for the analyser data.

However, the profile likelihood shown in Figure 2 is all but normal, the ordinate at the boundary $\sigma^2 = 0$ being 0.6 rather than 0. Therefore, we use a parametric bootstrap (10,000 simulations) to approximate to the distribution of $-2\log\mathcal{L}_{profile}(\sigma^2)$ – see Davison and Hinkley (1997, pp. 196–7). The 10,000 samples are generated in accordance with the structure (1) using the MLEs from (12) as the “true” parameter values. The upper 95% critical value from the simulation is 5.878, so that the profile likelihood confidence interval is

$$\{\sigma^2 : \mathcal{L}_{profile}(\sigma^2) > \exp(-5.878/2) = .053\} = (0, 0.029).$$

This confidence interval is 20% wider than that obtained in (13) from the asymptotic theory. We repeated the parametric bootstrap simulations using

two alternative sets of “true” parameter values, namely (a) the mean unbiased estimators (found by using a bootstrap estimate of bias)

$$\tilde{\sigma}^2 = 0.0066, \tilde{\boldsymbol{\theta}} = [0.1082 \ 0.1105 \ 1.5796] \text{ and } \tilde{\boldsymbol{\mu}} = [-0.01 \ -0.05 \ 26.13]$$

and (b) the median unbiased Grubbs estimators

$$\tilde{\sigma}^2 = 0.0129, \tilde{\boldsymbol{\theta}} = [0.1010 \ 0.1050 \ 1.6216] \text{ and } \tilde{\boldsymbol{\mu}} = [-0.01 \ -0.05 \ 26.13].$$

The upper 95% critical values from the simulations were 5.741, which leads again to a (0, 0.029) confidence interval, and 6.941 which gives (0, 0.031) as a confidence interval. Thus, the parametric bootstrap procedure seems in this particular instance not to be overly sensitive to the choice of “true” parameter values that are used in the simulation.

The same method can be applied to the likelihood (6) which comes from the reduced data (U_1, U_2) and for which the Grubbs estimators are the MLEs. Then the 95% profile likelihood confidence interval for σ^2 is found to be (0, 0.037), which is a substantially (19%) wider interval than that produced by the full likelihood.

In conclusion we note that situations where the product variance τ^2 is apparently zero, or very close to zero, do occur in practice. One example can be found in the data set appended to De Korte (2007). Here, coal samples were taken every five minutes over a period of about 2.5 hours. The product variance can be expected to be relatively small over such a short period. The three pairwise sample covariances, each of which is a moment estimate of τ^2 , are $S_{Y,X_1} = -0.0895$, $S_{Y,X_2} = -0.0123$ and $S_{X_1,X_2} = -0.0243$. These values suggests strongly that τ^2 might be zero. Indeed, the full data MLE of τ^2 is zero and the full data MLE of σ^2 is 0.337 with an estimated

standard error 0.102. The Grubbs estimate, on the other hand, is $\hat{\sigma}^2 = 0.427$ with an estimated standard error of 0.212, almost twice that of the full data MLE. The extent of the difference between the two standard error estimates is a manifestation of the substantial superiority, established in Section 3, of the full data MLE over the Grubbs estimator when τ^2 is close to zero.

5. Examining the assumptions

One area of concern regarding the applicability of the proposed method is the possibility that the daily coal quality values $\{T_i\}$ may exhibit serial correlation. Such correlation would certainly invalidate the assumption that the observation triples (Y_j, X_{j1}, X_{j2}) , $j \geq 1$ are independent. In this regard it should be noted that the Grubbs estimator (2) has the virtue that it is entirely unaffected by any such correlation because the T_j do not enter into its calculation at all. Thus, possible serial correlation is a matter of concern only in the application of the full data likelihood method. Since the daily instrument measurement errors are doubtlessly uncorrelated, it follows for instance that

$$\text{corr}(T_1, T_{1+k}) = \text{corr}(Y_1, Y_{1+k}) \times (1 + \sigma^2/\tau^2)$$

for $k \geq 1$ so that the existence of serial correlation among the T_j can be checked by examining the sample autocorrelations in the $\{Y_j\}$ series. In fact, since in our applications τ^2 is typically two orders of magnitude larger than σ^2 , we have $\text{corr}(Y_1, Y_{1+k}) \approx \text{corr}(T_1, T_{1+k})$. Figure 3 shows the correlogram of the analyser series in our first data set ($n = 94$) together with the customary confidence bounds at $\pm 2/n^{1/2} = \pm 0.206$. One point at lag $k = 3$ lies marginally outside the lower bound. Since there is no physical reality that would explain a negative autocorrelation at lag 3 in this particular coal quality process, we take it that an assumption of zero serial correlation is justified.

The normality assumption can be checked by examining Q-Q plots of the data produced by the analyser and the reference instruments. Neither Figure 4, which shows such a plot for the data produced by the analyser, nor any of the Q-Q plots (not shown here) corresponding to the other two reference instruments provide any basis for rejecting the normality assumption.

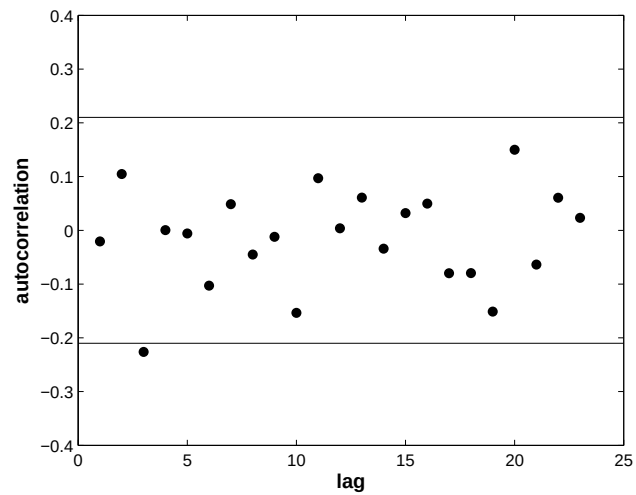


Figure 3. Correlogram of analyser data.

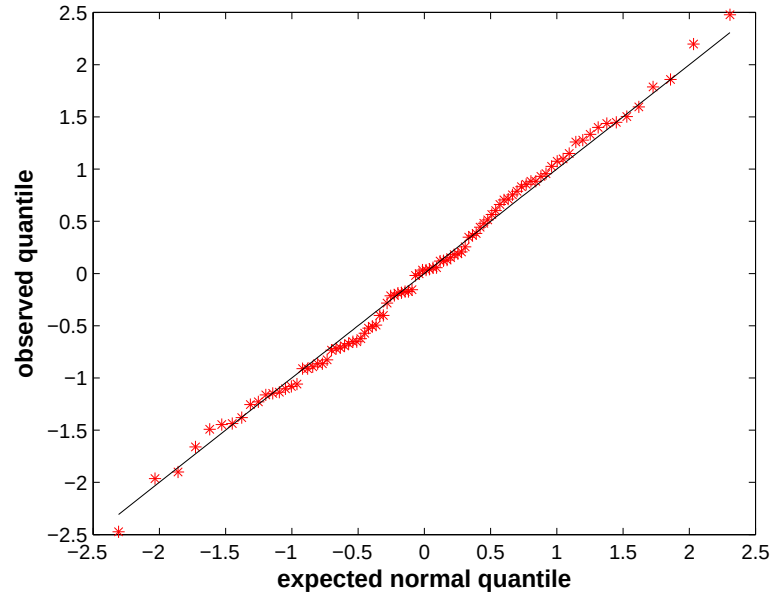


Figure 4. Q-Q plot of standardised analyser output data.

6. Summary

We have reconsidered a method proposed by Grubbs (1948) that can be used to estimate the error variance of a measuring instrument when replicate observations cannot be made but parallel measurements from other instruments are available. Our focus, motivated by a specific practical consideration, centres on the case where the error variances of the other instruments are greater by an order of magnitude of more than that of the instrument of interest. The Grubbs estimator has been compared with the MLE (maximum likelihood estimator). It has been shown that the relative efficiency of the MLE, which always exceeds 100%, is a decreasing function of the ratio of between samples variance to instrument variance. The conclusion is that if

the latter ratio is large, little is lost by using Grubbs estimator rather than the MLE. On the other hand, a great deal may be lost if Grubbs estimator is used and the product to analyser variance ratio is close to zero. However, obtaining confidence intervals based on the Grubbs method by the standard “estimate $\pm z_{1-\alpha} \times$ estimated standard error” formula is generally problematic unless the sample size is impractically large. As a result, we advocate use of a method based on the profile likelihood.

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7. Appendix

Proof of Proposition 1

Here, we set $\xi_1 = \sigma^2$, $\xi_2 = \sigma_1^2$, $\xi_3 = \sigma_2^2$, $\xi_4 = \tau^2$ and write Σ_{ξ} for $\Sigma_{\sigma^2, \theta}$. From (11), the log-likelihood function for the full data set is

$$l = \sum_{j=1}^n - \left\{ \log \det(\Sigma_{\xi}^{-1}) + (\mathbf{U}_j - \mu) \Sigma_{\xi}^{-1} (\mathbf{U}_j' - \mu') / 2 \right\}$$

Set

$$\mathcal{I}_{ij} = -E \left[\partial^2 l / \partial \xi_i \partial \xi_j \right].$$

We have

$$\mathcal{I}_{ij} = \text{tr} \left\{ \Sigma_{\xi}^{-1} P_i^{-1} \Sigma_{\xi}^{-1} P_j \right\} / 2 \quad (14)$$

with

$$P_i = \partial \Sigma_{\xi} / \partial \xi_i.$$

In the present setup, none of the P_i depend on ξ . Write

$$\Sigma_{\xi} = \begin{bmatrix} A & B \\ B' & D \end{bmatrix}$$

with

$$A = \begin{bmatrix} \sigma^2 + \sigma_1^2 & \sigma^2 \\ \cdot & \sigma^2 + \sigma_2^2 \end{bmatrix}, \quad B = \begin{bmatrix} -\sigma_1^2/2 \\ -\sigma_2^2/2 \end{bmatrix}; \quad D = \tau^2 + (\sigma_1^2 + \sigma_2^2)/4.$$

From 2.7 on page 33 of Rao (1973), for the full data $\mathbf{U} = (U_1, U_2, U_3)$,

$$\text{cov}(\mathbf{U})^{-1} = \Sigma_{\xi}^{-1} = \begin{bmatrix} A^{-1} + FF'/E & -F/E \\ -F'/E & 1/E \end{bmatrix} \quad (15)$$

where $E = D - B'A^{-1}B$ and $F = A^{-1}B$. Notice that E , FF'/E and F/E are all $O(\tau^{-2})$ as $\tau^2 \rightarrow \infty$. In fact,

$$E, FF'/E, F/E \downarrow 0 \text{ as } \tau^2 \uparrow \infty$$

so that

$$\Sigma_{\xi}^{-1} \downarrow \begin{bmatrix} A^{-1} & \mathbf{0} \\ \mathbf{0} & 0 \end{bmatrix}. \quad (16)$$

On the other hand, for the reduced data $\mathbf{U}^* = (U_1, U_2)$, we have

$$\text{cov}(\mathbf{U}^*) = A,$$

which, together with (16), proves (8).

If $\tau^2 \rightarrow 0$, we see that $E \rightarrow B'A^{-1}B$, so that

$$\Sigma_{\xi}^{-1} \rightarrow \begin{bmatrix} A^{-1} + FF'/B'A^{-1}B & -F/B'A^{-1}B \\ -F'/B'A^{-1}B & 1/B'A^{-1}B \end{bmatrix} := \Sigma^{-1}.$$

A detailed calculation using (14) with Σ_{ξ}^{-1} replaced by Σ^{-1} then leads to (9). Finally, (10) follows from (9) and (3).